

Artificial Intelligence (AI) is the field of creating systems capable of performing tasks that normally require human intelligence. AI includes reasoning, learning, perception, language understanding, decision-making, and problem solving. Modern AI systems are powered by machine learning and deep learning models that learn patterns from large amounts of data. Artificial Intelligence (AI) is the field of creating systems capable of performing tasks that normally require human intelligence. AI includes reasoning, learning, perception, language understanding, decision-making, and problem solving. Modern AI systems are powered by machine learning and deep learning models that learn patterns from large amounts of data. Artificial Intelligence (AI) is the field of creating systems capable of performing tasks that normally require human intelligence. AI includes reasoning, learning, perception, language understanding, decision-making, and problem solving. Modern AI systems are powered by machine learning and deep learning models that learn patterns from large amounts of data.

Machine Learning enables computers to learn from data without being explicitly programmed. Models identify patterns and make predictions. Supervised learning uses labeled data, unsupervised learning discovers hidden structures, and reinforcement learning learns through rewards and penalties. Machine Learning enables computers to learn from data without being explicitly programmed. Models identify patterns and make predictions. Supervised learning uses labeled data, unsupervised learning discovers hidden structures, and reinforcement learning learns through rewards and penalties. Machine Learning enables computers to learn from data without being explicitly programmed. Models identify patterns and make predictions. Supervised learning uses labeled data, unsupervised learning discovers hidden structures, and reinforcement learning learns through rewards and penalties.

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Models are neural networks trained on massive text corpora. They learn language patterns and can generate human-like responses. Examples include transformer-based architectures that understand context and predict the next token in a sequence. Large Language Models are neural networks trained on massive text corpora. They learn language patterns and can generate human-like responses. Examples include transformer-based architectures that understand context and predict the next token in a sequence.

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Vectorization converts unstructured data into mathematical representations. Text embeddings are one form of vectorization. These vectors allow machine learning systems to perform similarity calculations and semantic understanding. Vectorization converts unstructured data into mathematical representations. Text embeddings are one form of vectorization. These vectors allow machine learning systems to perform similarity calculations and semantic understanding. Vectorization converts unstructured data into mathematical representations. Text embeddings are one form of vectorization. These vectors allow machine learning systems to perform similarity calculations and semantic understanding.

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Chapter 7: Difference Between Tokens and Embeddings

Tokens represent pieces of text, while embeddings represent meaning. Tokenization breaks text apart. Embedding models convert those tokens into vectors that capture semantic relationships. Tokens represent pieces of text, while embeddings represent meaning. Tokenization breaks text apart. Embedding models convert those tokens into vectors that capture semantic relationships. Tokens represent pieces of text, while embeddings represent meaning. Tokenization breaks text apart. Embedding models convert those tokens into vectors that capture semantic relationships.

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Transformers revolutionized NLP through self-attention mechanisms. Unlike recurrent models, transformers process sequences in parallel and capture long-range dependencies efficiently.

how much information a model can consider at once. Larger context windows improve long-document understanding but increase computational costs.

solely on training data, the model retrieves relevant information from external knowledge sources before generating responses.

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improves factual accuracy, enables domain-specific knowledge integration, and keeps information up to date without retraining the entire model.

pipeline includes document ingestion, chunking, embedding generation, vector database storage, retrieval, reranking, prompt construction, and response generation.

recursive chunking, semantic chunking, sentence-based chunking, paragraph chunking, and hierarchical chunking.

AI assistant typically includes ingestion pipelines, embedding models, vector databases, retrieval systems, prompt templates, APIs, monitoring, and deployment infrastructure.

systems, autonomous agents, improved reasoning, efficient models, stronger governance, and widespread enterprise adoption.